

Diabetes Prediction Using Single Layer Perceptron

Simon B Siagian
The University of Adelaide
Adelaide, SA, Australia

simonbaharja.siagian@adelaide.edu.au

Abstract

Diagnosis of diabetes has been a major concern in the medical sector nowadays, as it has become one of the leading factors for mortality. In this study, we design a classification algorithm using Single Layer Perceptron (SLP) to predict if patients suffer from diabetes or not using several features in PIDD. We evaluate the model performance on two kinds of datasets, the original dataset and the dataset with replaced extreme outliers. We optimize the SLP by tuning several hyperparameters such as learning rate, activation function, and weight initialization. Result shows that the optimized SLP works better on data whose outliers are replaced with validation and test accuracy of 83.74% and 72.73% respectively.

1. Introduction

The medical sector has shown a great deal of machine learning applications. Some have been disease diagnosis and prediction, medical image recognition, and decision support system [1]. One of the most common diseases is diabetes, due to the high consumption of sugary product and the risk of this disease has increased worldwide [2].

In medical applications, machine learning algorithms generally utilize recorded data of former patients to find the pattern of symptoms and classify diseases. Some widely used machine learning algorithms are the neural network, decision tree, support vector machine, boosting algorithm, and logistic regression [3]. Sarwar et al. [4] used six ML algorithms and found that SVM and KNN performed better compared to other algorithms with accuracy of 77%. In [5], authors used various metrics to evaluate diabetes classification, such as accuracy, precision, recall, and f-score. For some tasks especially medical tasks, metrics weighted to minimize unwanted results such as misdiagnosis are more useful. Padmavathi [6] performed diabetes classification using several ML algorithms and found the optimal result with inter-quartile range data cleaning and standardization.

This study will focus on the application of the Perceptron Algorithm. Perceptron is one of the earliest developed classification algorithms that can be used in medical sector. This algorithm works very well with

linearly separable data [2]. This study uses and evaluates Single Layer Perceptron (SLP) performance to classify the risk of diabetes. Dataset used in this study is PIMA Indians Diabetes Dataset (PIDD) obtained from Kaggle [7]. This dataset contains 8 features with two labels: diabetes and non-diabetes.

2. Method

This part will explore the dataset, algorithm, and technique used in this study.

2.1. Dataset

The PIMA dataset of diabetes was downloaded from Kaggle [7] and was originally from the National Institute of Diabetes and Digestive and Kidney Diseases. This dataset includes 8 numerical variables about diabetes and one dependent variable ‘Outcome’ (diabetes or non-diabetes). The total sample of 768 was gathered from female patients with age of at least 21 years old. Table 1 show the description of features in PIMA dataset used in this study.

2.2. Perceptron

Perceptron is an algorithm used for binary classification. This algorithm is supervised, meaning it needs ground truth data in its learning process. Perceptron is a type of linear classifier, which makes its prediction based on a linear combination of predictors or dependent variables [2].

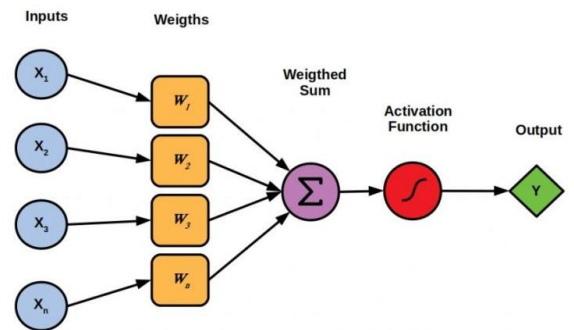


Figure 1. Architecture of SLP

Table 1. Description of PIMA Indian Diabetes Dataset Features

Attributes	Description	Range	Data Type	# Missing values
Pregnancies	Number of times pregnant	0-17	Numerical	0
Glucose	Plasma glucose concentration from 2 hours oral glucose tolerance test	0-199	Numerical	5 (0.6%)
Blood Pressure	Diastolic blood pressure in mmHg	0-122	Numerical	35 (4.55%)
Skin Thickness	Triceps skin fold thickness (mm)	0-99	Numerical	227 (29.55%)
Insulin	Two-hour serum insulin (mu U/ml)	0-846	Numerical	374 (48.69%)
BMI	Body mass index (kg/m)	0-67.1	Numerical	11 (1.43%)
Diabetes Pedigree Function	Diabetes history in relatives and genetic relationship to the subject	0.078 – 2.42	Numerical	0
Age	Age of subjects	21-81	Numerical	0

Perceptron distinguishes class with linear decision boundary. This algorithm takes a set of scalar input features with a constant ‘bias’ and assign weight to each feature.

The result of this linear combination is fed through an activation function to determine the class. The original perceptron Rossenbalt’s perceptron (1957) can be defined as follows:

$$h(z) = \begin{cases} 0 & \text{if } z < 0 \\ 1 & \text{if } z \geq 0 \end{cases}$$

Another activation function that is useful to introduce nonlinearity to the model and commonly used in neural network is sigmoid activation function, this function allows gradient descent to work [5], described as follows:

$$\phi(z) = \frac{1}{1 + e^{-z}}$$

The linear combination of features is shown in the following equation:

$$z = \mathbf{x}^T \mathbf{w} + b \quad (1)$$

Rossenbalt’s Perceptron learns by minimizing prediction error generated when it attempts to classify input data. The parameter (weight) will be updated if there is an incorrect classification. The weight will be adjusted repeatedly until the prediction is true. Mathematically:

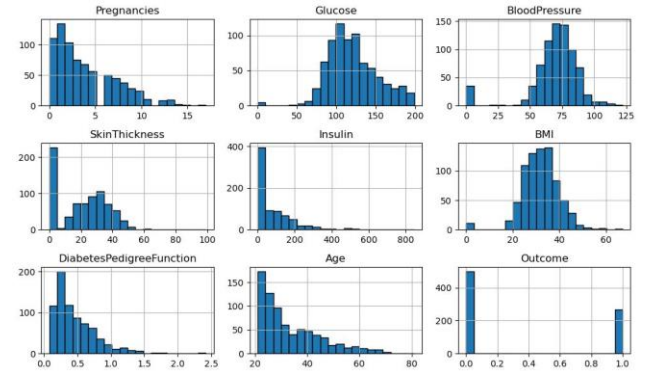
$$w_i^{n+1} = w_i^n + v (y - \hat{y}) x_i \quad (2)$$

Where v is the learning rate, and $(y - \hat{y})$ is prediction error.

2.3. Data Exploration

Obtaining a good understanding of the data is a necessary task in machine learning. PIMA Indian Diabetes Dataset (PIDDD) contains 768 observations, and visual tools is used to visualize the distribution of all attributes in this dataset. Figure 2 shows the distribution of the data. Some

biological features are right skewed because the dominant 0 values (i.e. Skin Thickness and Insulin). In this dataset, biological parameters such as blood pressure, skin thickness, insulin, and glucose have zero values which are inaccurate according to domain knowledge [8], thus it seems that null or *NaN* values have been coded as zero.

**Figure 2. Distribution of data in PIDDD**

However, removing *NaN* values will significantly reduce the number of samples, and avoid removing rows is preferred. On the other hand, imputing all *NaN* values with mean or median will introduce bias to the data. Hence, in this study imputer is only performed for variables with the least zero values to retain the originality of the data and at the same time keep the amount of data [9]. The label in this dataset is not severely unbalanced with around 35% positive class, thus it is appropriate to use this data for modeling without further data augmentation or synthesis.

However, to retain the proportion of each class in data splitting, stratified split is used. There will be two forms of dataset used in this experiment. First is the original data with some of NaN in features with the least missing values are imputed. This is to retain the number of data and keep the useful information in the data as imputing a small amount of data has a smaller impact on the dataset [6]. Second is dataset with outlier has been replaced using 5th and 95th percentile method. This is to reduce the impact of extreme values and skew the model.

Subsequently, data are normalized using a standard scaler to make all the features uniform. This normalization method is also less sensitive to outliers compared to methods such as MinMax scaler.

2.4. Modelling and Evaluation

SLP will be applied from scratch using *pytorch*. Step and sigmoid activation function will be used for the perceptron. Each model or hyperparameter combination will be evaluated with metrics such as accuracy, balanced accuracy, f1 score [6]. Accuracy is used to measure the total ratio of correct predictions made by the model, described as follows:

$$Acc = \frac{TP + TN}{TP + FP + FN + TN}$$

As the label is imbalanced, there is a high chance for high conventional accuracy. Thus, balanced accuracy can be used to evaluate the performance of model for imbalanced dataset as it gives equal weight to both minority and majority class, described as follows:

$$Balanced\ Acc = \frac{Sensitivity + Specificity}{2}$$

Balanced accuracy is also useful when precision and recall are critical, especially in medical diagnosis [10]. For an imbalance dataset, especially in a low positive class, f1 measure is also useful as it focuses on how well the model performs on the positive class [6]. The formula for f1 score is described as follows:

$$F - Measure = \frac{2 * Precision * Recall}{Precision + Recall}$$

3. Result and Analysis

The block diagram of experiment in this study is shown in Figure 3. There are two datasets to be evaluated: the original data and the data whose outliers have been replaced using percentile data, this is to evaluate how data preprocessing affects our prediction model. Data was then divided using the fix split method into three splits: train,

validation and test set with proportions of 64%, 16% and 20% respectively. After splitting, data was preprocessed using pipeline to avoid data leaking.

SLP is built using object orientation programming with *pytorch*, to allow optimizing the model by override the inherited class method.

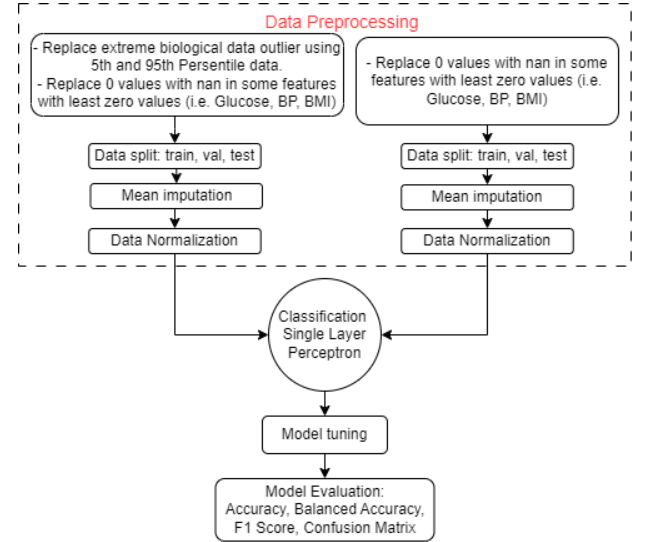


Figure 3. Block diagram of the experiment

There are several steps of model tuning or optimization done in this experiment. Described as follows:

- Stage 1: baseline model, SLP with step activation function and zero weight initialization.
- Stage 2a, 2b, 2c: the activation function is replaced with sigmoid and uses zero, random and *glorot* initialization respectively.
- Stage 3: model is evaluated on different learning rate to see the performance based on the learning rate.
- Stage 4: model with regularization or penalty is evaluated.

In each stage, model is evaluated with metrics accuracy, balanced accuracy and f-score. In model optimization, all decisions are based on performance on the validation set by putting more concern in balanced accuracy and f1 score.

Figure 4 shows the model performance comparison for every stage. In this experiment, all stages are done with 500 iterations, this number is chosen because, from a number of trials, the parameters are converging around 500 iterations. In each stage, the model performance improved in all metrics indicating that model tuning works.

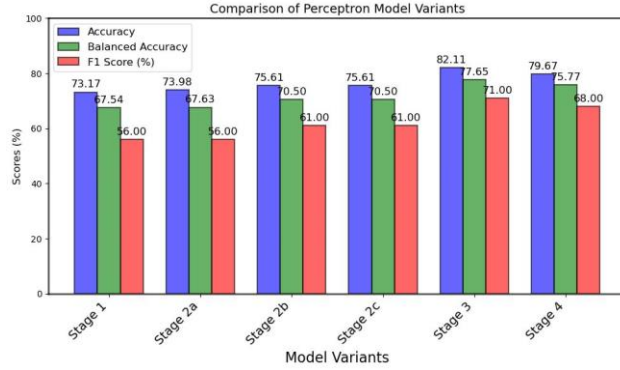


Figure 4. Comparison in model optimization

Based on the experiment, using sigmoid function slightly improved the model. This is due to the nature of sigmoid function that allows perceptron to learn the relationship between inputs and outputs where they are not strictly linear [6]. Moreover, using random or glorot initialization produces better result than zero initialization, as random and glorot initialization allows the perceptron to assign different importance to different input features during training.

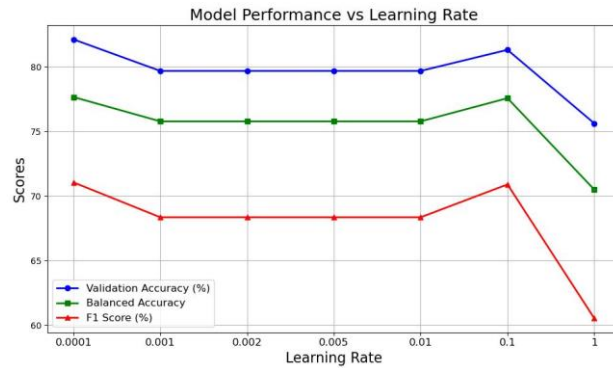


Figure 5. Model performance on various learning rate

To obtain the best learning rate for perceptron, the model was trained using several learning rates and tested against validation set. Figure 5 shows that learning rate = 10^{-4} results in the best performance for all metrics with the highest balanced accuracy at 77.65% and f1 score 0.72. In the last stage, model was trained with penalty or regularization terms to improve the generalization ability. However, from several number of trials, model with penalty produces lower performance than the previous one. Thus, for this experiment, based on the performance on validation set, the best model is chosen at stage 3 with a learning rate = 10^{-4} .

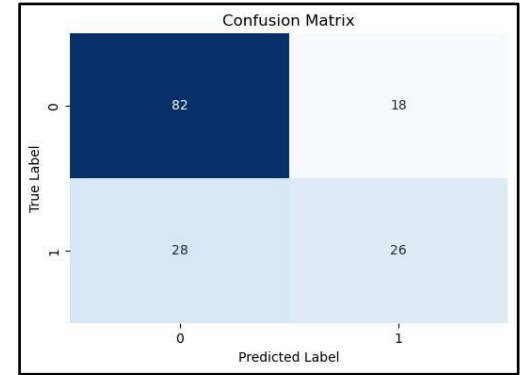
The best model's hyperparameter obtained from stage 3 (training with original data) then is trained and tested on the data with outliers replaced. Table 2 shows the

comparison of the model with the best hyperparameter on both dataset.

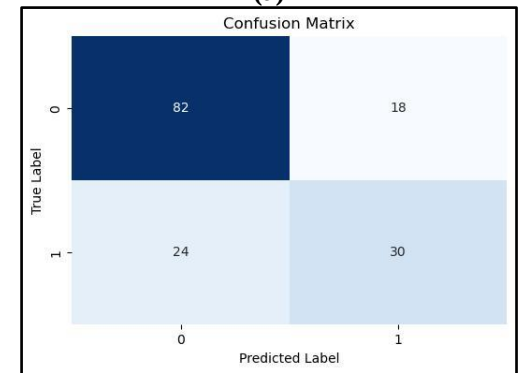
Table 2. Best Performance on Validation Set

Dataset	Best Acc (%)	Best Balanced. Acc (%)	Best F1-score
Original	82.93	78	0.72
Replaced Outlier	83.74	80	0.74

SLP works better on data with outlier replaced, with higher performance in all metrics. The model for both datasets was tested on test set to compare the generalization ability. Performance on test set for each dataset is shown by the confusion matrix in Figure 6. The random state for each data shuffle is the same to ensure that each data split has the same data point, except for the outliers that have been replaced.



(a)



(b)

Figure 6. Best model's confusion matrix on test set (a) Original data (b) Outlier replaced

Models generalize better on dataset with outlier replaced with accuracy, balanced accuracy and f1 score is 72.73%, 69% and 0.6 respectively. Model on data with outlier replaced also predict lower False Negative Rate (FNR) which is crucial in medical diagnosis.

Overall, the perceptron classifier does not work very well in classifying diabetes data with maximum validation accuracy is 83.74% on validation set and 72.73% on test set with a relatively high false negative rate (around 44%) as this dataset is linearly inseparable [1], meaning that the dataset has a complex relationship between features and output. Meanwhile, SLP works by defining hyperplane or linear boundary between classes, thus this algorithm struggles to discriminate the class.

4. Conclusion and Future Work

This study aims to design a simple perceptron algorithm to classify patients with diabetes in PIDD. Before algorithm modeling, several preprocessing steps are done as PIDD contains missing and noisy data which can negatively impact the performance of the classifier. To analyze the effect of the dataset on the classifier algorithm, we build and test the model on two datasets: the original dataset, and the dataset with outlier replaced with 5th and 95th percentiles. However, only the first dataset is used for model optimization using a fixed validation set. Several model's hyperparameters were tuned to find the best combination such as activation function, initialization, learning rate, and penalty or regularization. The best SLP is found with the combination of sigmoid activation function, random or glorot initialization, and learning rate 10^{-4} . Results show that SLP works better on cleaned data where extreme outliers are replaced or adjusted with 5th and 95th percentiles with balanced accuracy and f1 scores are 83.74% and 0.74 respectively. Replacing outliers helps reduce the skewness of the model on a dataset that already contain many zero or null values.

Higher performance is expected when using nonlinear algorithm such as random forest tree or more advanced and complex classification algorithm such as Multi-Layer Perceptron (MLP) with backpropagation. A dataset with more samples and clean data is also necessary for model building in the future.

5. Code

All code and datasets used in this experiment are available at <https://github.com/simon03instru/diabetes-perceptron>. This repository contains a separate notebook for modeling and data exploration.

References

- [1] Bukhari, M.M., et al., *An improved artificial neural network model for effective diabetes prediction*. Complexity, 2021. **2021**(1).
- [2] Mirshahvalad, R. and N.A. Zanjani, *Diabetes prediction using ensemble perceptron algorithm*. 2017 9th International Conference on Computational Intelligence

- and Communication Networks (CICN), IEEE, 2017: p. 190-194.
- [3] Dweekat, O.Y. and S.S. Lam, *Optimized design of hybrid genetic algorithm with multilayer perceptron to predict patients with diabetes*. Soft Computing (Berlin, Germany), 2023. **27**(10): p. 6205-6222.
- [4] Sarwar, A., et al., *Diagnosis of diabetes type-II using hybrid machine learning based ensemble model*. International Journal of Information Technology, 2020. **12**: p. 419-418.
- [5] Chris. *Why you can't truly create Rosenblatt's Perceptron with Keras*. 2019 [cited 2024 15-09]; Available from: <https://machinecurve.com/index.php/2019/07/24/why-you-cant-truly-create-rosenblatts-perceptron-with-keras>.
- [6] Padmavathi, M.S. and C.P. Sumathi, *A new method of data preparation for classifying diabetes dataset*. Indian Journal of Science and Technology, 2019. **12**(22).
- [7] LEARNING, U.M. *Pima Indians Diabetes Database*. 2016 [cited 2024 15-09]; Available from: <https://www.kaggle.com/datasets/uciml/pima-indians-diabetes-database>.
- [8] UA, Z. and K. N., *Predicting diabetes in medical datasets using machine learning techniques*. Int J Sci Eng Res, 2017. **5**(2): p. 257-267.
- [9] Ram, A., *A Density based algorithm for discovering density varied clusters in large spatial databases*. International Journal of Computer Applications **3**(6).
- [10] Fallucchi, F. and A. Cabroni, *Predicting risk of diabetes using a model based on multilayer perceptron and features extraction*. Journal of Computer Science, 2021. **17**(9): p. 748-761.
- [11] QUANSTART. *Introduction to Artificial Neural Networks and the Perceptron*. 2024 [cited 2024 15-09]; Available from: <https://www.quantstart.com/articles/introduction-to-artificial-neural-networks-and-the-perceptron/>.